Fundamentals 27

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Problem 27.1

(During the problem session it was clarified that the problem meant to prove the other direction of Urysohn's Lemma)

Assume that for any two nonempty disjoint closed $A, B \subseteq X$, there is $f : X \to [0,1]$ such that $f(A) = \{0\}$ and $f(B) = \{1\}$. We want to show that $\mathbf{T_4}$. Let A, B be closed and disjoint. Then there exists continuous f such that $f(A) = \{0\}$ and $f(B) = \{1\}$. Let

$$C = f^{-1}\left(\left[0, \frac{1}{2}\right)\right), \quad D = f^{-1}\left(\left(\frac{1}{2}, 1\right]\right).$$

We have that $A \subseteq C$, $B \subseteq D$ and since $[0, 1/2)$ and $(1/2, 1]$ are open in $[0,1]$, C, D are continuous and disjoint. Therefore X is $\mathbf{T_4}$.

Problem 27.2

Let $x \in \mathbb{R}$ and $A \subseteq X$ be closed and not containing x. Let

$$y_1 = \sup\{y \in A : y < x\}, \text{ or } -\infty,$$

$$y_2 = \inf\{y \in A : y > x\}, \text{ or } \infty.$$

Since A is closed, if $y_1, y_2 \neq -\infty, \infty$, then $y_1, y_2 \in A$ because closed sets contain their limit points.

We can let

$$f(y) = \max\left(0, 1 - \frac{|x-y|}{\min(|x-y_1|, |x-y_2|)}\right)$$

whenever $y \in (y_1, y_2)$ (creating a spike around x and within (y_1, y_2)) and $f(y) = 0$ everywhere else, so that f is continuous, $f(A) = \{0\}$, and $f(x) = 1$.

Problem 27.3

Let $A \subseteq Y$ be closed and pick $y_0 \in Y \backslash A$. We know that A is closed if there exists some closed $C \subseteq X$ with $A = C \cap Y$. Since X is completely regular, there is a continuous $f : X \to [0, 1]$ with $f(y_0) = 1$ and $f(C) = \{0\}$. Restricting f to Y, we see that $f|_Y(y_0) = 1$ and $f|_Y(A) = f(C \cap Y) = \{0\}$, so Y is completely regular.

Problem 27.4

Part (1)

Assume $x \in A$. Then one of the distances in $\{d(x, y) : y \in A\}$ is $d(x, x) = 0$, so $\inf\{d(x, y) : y \in A\} = 0$.

Assume $\inf\{d(x, y) : y \in A\} = 0$. For each integer $n \geq 1$, there is some $y_n \in A$ with $d(x, y_n) < \frac{1}{n}$. Therefore, the sequence (y_n) satisfies $d(x, y_n) \to 0$, so $y_n \to x$. But A is closed, so it contains its limit points, so $x \in A$.

Part (2)

Let $A, B \subseteq X$ be disjoint and closed. By part (1),

$$D(x, A) = 0 \iff x \in A, \quad D(x, B) = 0 \iff x \in B.$$

Since A and B are disjoint, for all $x \in X$, at least one of $D(x, A), D(x, B)$ is positive, so

$$D(x, A) + D(x, B) > 0$$

for all $x \in X$. Therefore, the function

$$f(x) = \frac{D(x, A)}{D(x, A) + D(x, B)}$$

is well-defined and takes values in $[0, 1]$.

If $x \in A$, then $D(x, A) = 0$ and $D(x, B) > 0$ so $f(x) = 0$. If $x \in B$, then $D(x, B) = 0$ and $D(x, A) > 0$, so $f(x) = \frac{D(x,A)}{D(x,A)+0} = 1$. So it remains to show that f is continuous.

We claim that each map $D(x, A)$ and $D(x, B)$ is Lipschitz, and therefore continuous. For any $x, y \in X$ and $a \in A$,

$$d(x, a) \leq d(x, y) + d(y, a) \implies d(x, a) - d(y, a) \leq d(x, y).$$

Taking the infimum over $a \in A$, we get

$$D(x, A) - D(y, A) \leq d(x, y).$$

By the same argument,

$$D(y, A) - D(x, A) \leq d(x, y).$$

Therefore, $|D(x, A) - D(y, A)| \leq d(x, y)$, so the function mapping x to $D(x, A)$ is Lipschitz. Similarly, the function $D(x, B)$ is Lipschitz, so $g(x) = D(x, A) + D(x, B)$ is Lipschitz as well.

Now, since $f(x) = D(x, A)/g(x)$ is the quotient of two continuous functions where the denominator is never 0, f is continuous.

Problem 27.9

Let $\{f_i : X \to [0, 1]\}_{i \in I}$ be a partition of unity on X subordinate to $\{U_i\}_{i \in I}$ and $\{g_i : X \to [0, 1]\}_{j \in J}$ be a partition of unity on Y subordinate to $\{V_j\}_{j \in J}$.

For each $(i, j) \in I \times J$ define

$$h_{(i,j)} : X \times Y \to [0, 1], \quad h_{(i,j)}(x, y) = f_i(x) g_j(y).$$

We have that

$$\operatorname{supp}(h_{(i,j)}) = \overline{\{x : f_i(x) \neq 0\} \times \{y : g_j(y) \neq 0\}}$$

$$= \overline{\{x : f_i(x) \neq 0\}} \times \overline{\{y : g_j(y) \neq 0\}} = \operatorname{supp}(f_i) \times \operatorname{supp}(g_j) \subseteq U_i \times V_j.$$

Also, for every $(x, y) \in X \times Y$,

$$\sum_{(i,j) \in I \times J} h_{(i,j)}(x, y) = \sum_{i \in I} \sum_{j \in J} f_i(x) g_j(y) = \left(\sum_{i \in I} f_i(x) \right) \left(\sum_{j \in J} g_i(y) \right) = 1 \cdot 1 = 1.$$

So $h_{(i,j)}$ is a partition of unity on $X \times Y$.